
Global Convergence and High-Frequency Spectral Bias of Low-Rank Networks

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Abstract

1 Low-rank neural networks are usually introduced as compressed versions of dense
2 models, but this view misses their main theoretical role: rank creates a structured
3 mean-field system whose optimization and spectral behavior can be analyzed
4 directly. We prove that low-rank random-feature neural networks, with frozen
5 feature maps and trainable channel weights, have well-posed mean-field dynamics
6 and that every convergent trajectory reaches a global minimizer of the population
7 risk under a standard loss-identifiability condition. The result holds for arbitrary
8 depth and standard independent initialization; low rank changes constants through
9 the channel mixing norm but does not destroy the dense-span argument needed
10 for global optimality. We then explain why convergence alone is not enough for
11 highly oscillatory functions. Building on the spectral-bias perspective of Rahaman
12 et al. [17], low rank is shown to control path-space degrees of freedom, continuous
13 piecewise affine geometry, and finite-time Fourier recovery. Experiments on
14 oscillatory targets and Fourier-native rank sweeps support the view that the useful
15 rank is often intermediate: small enough to change spectral transfer, but large
16 enough to avoid an approximation bottleneck.

17 1 Introduction

18 Low-rank neural networks replace dense matrices by factored or bottlenecked operators. The standard
19 motivation is computational: fewer trainable parameters and cheaper memory traffic. This paper
20 argues that the more important question is mathematical. Does the low-rank constraint preserve the
21 optimization guarantees available in mean-field theory, and can the same constraint help with the
22 spectral bias that prevents neural networks from learning highly oscillatory functions?

23 We answer the first question positively for a low-rank random-feature architecture. The model
24 freezes a rich random-feature map and trains low-rank channel weights through mean-field gradient
25 flow. Since the frozen features keep a dense span throughout training, the usual obstruction in deep
26 mean-field convergence proofs is removed: at a limit point, zero gradient implies the conditional
27 first-order condition for the loss, and the loss-identifiability assumption forces global optimality. Low
28 rank enters through the mixing matrix and channel index, changing constants but not the logic of the
29 proof.

30 The second question is more subtle. A globally convergent training dynamic can still learn low
31 frequencies first and fail at finite time on highly oscillatory targets. We therefore connect the
32 convergence theorem to a rank-dependent spectral bias mechanism. For ReLU networks, low
33 rank constrains active path coefficient tensors; this constrains switching normals, suppresses high-
34 codimension faces in the continuous piecewise affine complex, and changes Fourier shell behavior.
35 In one dimension, the raw spline tail remains close to the classical k^{-4} law, so the correct diagnostic
36 is not the asymptotic exponent alone but the learned Fourier recovery curve. This leads to a practical

rank-selection rule: choose the smallest rank that avoids the approximation bottleneck while flattening the finite-time spectral transfer.

Contributions. The paper makes three contributions:

- We state a global convergence theorem for arbitrary-depth low-rank random-feature networks in the mean-field limit.
- We turn low-rank spectral bias into a geometric statement: rank constrains active path coefficients, which constrains CPWL switching geometry and Fourier shell energy.
- We connect this mechanism to experiments on highly oscillatory targets, including high-frequency regression, frozen-vs-trainable random features, and heavy-tail spectral updates.

2 Related Work

Mean-field convergence. Mean-field theory analyzes neural network training through the deterministic evolution of parameter distributions in the large-width limit. Foundational work established global convergence mechanisms for two-layer models and Wasserstein gradient flows under suitable convexity, homogeneity, or support assumptions [13, 5]. Later multilayer mean-field analyses clarified how deep feature maps evolve and when stationary points can be related to global optima [16, 15]. Our result follows this line but focuses on low-rank random-feature networks: the existence and uniqueness estimates use the same mean-field template, while the convergence proof must be adapted to channel-wise low-rank couplings.

Low-rank and random features. Low-rank neural networks are often motivated by parameter efficiency, with LoRA-style decompositions now standard in large-scale adaptation [12]. Most theory for low-rank networks focuses on expressivity, approximation, or kernel behavior rather than global training dynamics. Random features provide a tractable bridge between finite networks and kernel methods: fixing the feature map turns the representation into a stable basis, while training the channel weights retains a mean-field optimization problem. In our RF-LR setting, frozen random features prevent support collapse and preserve the dense-span property needed for the stationarity-to-optimality argument; the low-rank factors enter through bounded channel mixing rather than through a full dense contraction chain.

Bottleneck rank and implicit low-rank bias. Recent work gives a functional notion of optimal rank that is closer to our setting than the algebraic rank of a single matrix. Jacot [9] studies the implicit bias of large-depth homogeneous networks and shows that their representation cost converges toward a nonlinear rank notion, sandwiched between Jacobian rank and bottleneck rank. The bottleneck rank is the smallest inner dimension k for which a function can be factored as $f = h \circ g$ through $\mathbb{R}^k$. Follow-up work [10] proves that learned deep representations exhibit a bottleneck structure and introduces finite-depth corrections that balance low inner dimension against regularity. This suggests a theoretical target rank: the smallest rank that captures the intrinsic bottleneck dimension of the target without forcing an irregular or rank-underestimating interpolant. Complementarily, Bantzis, Simon, and Jacot [3] show that the first saddle escape of deep ReLU networks has a low-rank bias in deeper layers and propose saddle-to-saddle dynamics with increasing bottleneck rank. Our rank-selection principle is consistent with this picture: training should start from low effective rank, but useful learning requires increasing rank until approximation and spectral recovery saturate.

Spectral bias. The spectral-bias phenomenon was formalized by Rahaman et al. [17], who showed that neural networks tend to learn low-frequency components before high-frequency components and related this behavior to Fourier decay properties of piecewise linear networks. This paper uses that viewpoint but changes the architectural knob: instead of only observing that high frequencies are learned late, we ask how rank changes the CPWL geometry that controls the finite-resolution Fourier transfer. The key difference is coefficient-level: low rank suppresses high-codimension face moments in the Fourier shell expansion, moving the observed spectrum toward the facet-dominated regime discussed by Rahaman et al. [17].

111 and the second-layer preactivation is

$$H_2(t, c_2; x) = \sum_{k=1}^r L_{c_2, k} f_k(t, x). \quad (3)$$

112 Thus H_2 is the reconstruction of the second-layer signal from a bounded rank- r channel basis. The
 113 output is obtained by averaging the trainable top weights against $\varphi_2(H_2)$. For deeper networks,
 114 the same alternating pattern is used: trainable channel weights are averaged against frozen random
 115 features and mixed through bounded low-rank channel matrices.

116 The population objective is

$$\mathcal{L}(W) = \mathbb{E}_{(X, Y)} [\mathcal{L}(Y, \hat{y}(X; W))]. \quad (4)$$

117 The mean-field gradient flow evolves the trainable weights by

$$\partial_t w_\ell(t, \cdot) = -\xi_\ell(t) \nabla_{w_\ell} \mathcal{L}(W(t)), \quad (5)$$

118 where ξ_ℓ are bounded learning-rate schedules. The exact formulas are standard backpropagation in
 119 the mean-field variables; the only low-rank modification is the channel mixing through $L_{c_\ell, k}$.

120 Under the technical regularity, diversity, loss-identifiability, and convergence assumptions stated in
 121 Appendix A, freezing the feature directions gives a well-defined mean-field system and allows the
 122 usual stationarity argument to conclude global optimality.

123 **Theorem 3.1** (Well-posedness and global convergence of RF-LR training). *Under Assumptions A.1–*
 124 *A.6, the low-rank mean-field gradient-flow system has a unique solution on every finite interval $[0, T]$*
 125 *and therefore on $[0, \infty)$. Moreover, if the resulting low-rank mean-field dynamics converges to W^* ,*
 126 *then W^* is a global minimizer of the population loss:*

$$\mathcal{L}(W^*) = \inf_W \mathcal{L}(W). \quad (6)$$

127 For every depth $L \geq 2$, this holds with standard independent initialization of the trainable weights.

128 The well-posedness part only justifies that the limiting dynamics are genuine ODEs rather than formal
 129 flows. The technical fixed-point estimates are standard mean-field estimates with the dense-layer
 130 constants replaced by low-rank mixing norms, and are not the focus of the paper.

131 *Proof idea.* The existence and uniqueness part follows the standard mean-field fixed-point argument.
 132 We only summarize the convergence-to-global-minimum step here; Appendix B gives the full notation
 133 and the five formal steps. We assume that gradient descent, or its mean-field gradient-flow limit, has
 134 converged somewhere: $W(t) \rightarrow \bar{W}$ in the coupling topology. Stationarity and the persistent dense
 135 span of the frozen first-layer features then imply the integrated identity

$$\mathbb{E}[d_L(Z; \bar{W}) \mid X = x] B_k^{(2)}(x; \bar{W}) = 0 \quad \text{for } \mathbb{P}_X\text{-a.e. } x, \quad k = 1, \dots, r, \quad (7)$$

136 where $d_L(Z; \bar{W}) = \partial_2 \mathcal{L}(Y, \hat{y}(X; \bar{W}))$ and $B_k^{(2)}$ is the channel-wise backpropagated factor defined
 137 in Appendix B.

138 *Step 3: remove the backpropagated factor.* The non-degeneracy assumptions exclude a dead-channel
 139 limit: for $\mathbb{P}_X$ -a.e. x , at least one channel has a nonzero backpropagated factor $B_k^{(2)}(x; \bar{W})$, and
 140 the analogous statement holds layer by layer for $L > 3$. For ReLU-type activations, the forward
 141 estimates only need the channel Lipschitz inequality

$$\left| \varphi\left(\sum_{k=1}^r x_k\right) - \varphi\left(\sum_{k=1}^r y_k\right) \right| \leq \sum_{k=1}^r |x_k - y_k|. \quad (8)$$

142 Consequently (7) implies

$$\mathbb{E}[\partial_2 \mathcal{L}(Y, \hat{y}(X; \bar{W})) \mid X = x] = 0 \quad \text{for } \mathbb{P}_X\text{-a.e. } x. \quad (9)$$

143 *Step 5: the trajectory loss converges to the limit loss.* Since $W(t) \rightarrow \bar{W}$ in the coupling topology,
 144 there are couplings π_t between $W(t)$ and $\bar{W}$ whose weighted channel gaps vanish. In the two-layer
 145 channel notation, the key convergence statement is exactly

$$\mathbb{E}_{\pi_t} \left[(1 + |\bar{w}_2|) |\bar{w}_2| \sum_{k=1}^r |\bar{w}_{1, k}| |w_{1, k} - \bar{w}_{1, k}| \right] \rightarrow 0, \quad (10)$$

with analogous quantities for all deeper layers. The low-rank form $H_2 = \sum_k L_{c_2,k} f_k$ bounds every forward and backward difference by these same channel integrals. Hence

$$\mathbb{E} [|\hat{y}(X; W(t)) - \hat{y}(X; \bar{W})|] \rightarrow 0. \quad (11)$$

Lipschitzness of the loss in its second argument on the a priori bounded trajectory yields

$$|\mathcal{L}(W(t)) - \mathcal{L}(\bar{W})| \leq K \mathbb{E} [|\hat{y}(X; W(t)) - \hat{y}(X; \bar{W})|] \rightarrow 0. \quad (12)$$

Step 4: identify the limit as globally optimal. By the loss-identifiability condition,

$$\mathbb{E}[\partial_2 \mathcal{L}(Y, u) \mid X = x] = 0 \implies \mathbb{E}[\mathcal{L}(Y, u) \mid X = x] = 0. \quad (13)$$

Applying this with $u = \hat{y}(x; \bar{W})$ and using non-negativity of $\mathcal{L}$ gives

$$\mathcal{L}(\bar{W}) = 0 = \inf_W \mathcal{L}(W). \quad (14)$$

Together with Step 5, $\mathcal{L}(W(t)) \rightarrow \mathcal{L}(\bar{W})$. $\square$

4 Low-Rank Spectral Bias

Related work: spectral bias. Theorem 3.1 says that any convergent trajectory reaches a global minimizer. It does not say how fast different frequencies are learned, nor which finite-time solution is reached under a fixed training budget. Highly oscillatory functions expose this gap. This is precisely the phenomenon called spectral bias by Rahaman et al. [17]: standard neural networks tend to fit low-frequency components before high-frequency components. Rahaman et al. [17] explain this behavior through Fourier decay and training dynamics of neural networks.

Our mechanism is complementary and more structural. In ReLU networks, the represented function is continuous piecewise affine (CPWL), so Fourier behavior can be related to the polyhedral complex induced by activation boundaries. Low rank does not merely reduce parameter count; it constrains the algebraic degrees of freedom of the active path coefficients, and this changes which CPWL faces can carry Fourier mass. Our goal is therefore not to contradict classical spectral bias, but to identify rank as an architectural knob that modifies the geometry behind the finite-time Fourier transfer.

In a ReLU network, on every activation region the network is affine, and the affine coefficient is a sum over active paths. The high-level idea is that a low-rank network has fewer *independent* paths in this sum-over-paths representation. More precisely, when $W_\ell = U_\ell V_\ell^\top$ with rank r_ℓ , the path coefficients involving layer ℓ must factor through an r_ℓ -dimensional latent coordinate. Thus the graph may still contain many symbolic paths, but the number of independent path constraints is much smaller than in a full-width layer. Fewer independent path constraints mean fewer independent switching constraints for the polytopes of the CPWL linear-region complex. Geometrically, this favors more high-dimensional faces and suppresses low-dimensional, high-codimension intersections. Spectrally, the Fourier shell law below then predicts a lower finite-resolution decay, closer to the facet-dominated regime.

Mechanism summary. The chain of implications is the following. A factorization $W_\ell = U_\ell V_\ell^\top$ does not only compress parameters; it forces all layer- ℓ directions to pass through an r_ℓ -dimensional bottleneck. This reduces the number of independent path coefficients, which reduces the number of independent switching normals that can cut the input space. The CPWL complex then contains fewer stable high-codimension intersections and relatively more facet-dominated contributions. Since the Fourier expansion of a CPWL function is organized by codimension, suppressing high-codimension face moments moves the finite-resolution spectrum toward the slow-decay facet regime described by Rahaman et al. [17]. The useful rank is therefore not the smallest rank: if r is too small, approximation error dominates; if r is too large, the network recovers the full-rank spectral bias. The target is the first intermediate rank where approximation is controlled and the Fourier recovery scaling has already saturated.

This gives a theoretical way to define an optimal rank. At the population level, the natural lower bound is the bottleneck rank of the target, $\text{Rank}_{BN}(f^*)$ [9, 10]. Under finite training time and finite data, the practical rank should minimize a three-term trade-off

$$r^*(T, \varepsilon) \in \arg \min_r \{ \mathcal{A}_r(f^*) + \lambda_T \mathcal{O}_r(T) + \lambda_F \mathcal{S}_r \}, \quad (15)$$

where $\mathcal{A}_r(f^*)$ is the rank- r approximation error, $\mathcal{O}_r(T)$ is an optimization cost at time T , and $\mathcal{S}_r$ is the spectral-bias cost measured by Fourier recovery or CPWL face moments. In experiments, this criterion is estimated by the first rank for which approximation error is small and the log-log Fourier recovery slope no longer improves. Thus the predicted optimum is a saturation rank, not full rank.

This algebraic constraint becomes geometric. Inside a cell of the previous layers, a new switching normal has the form

$$n_{\ell i \varepsilon} = B_{\ell-1, \varepsilon}^\top V_\ell u_{\ell i} \in \text{Im}(B_{\ell-1, \varepsilon}^\top V_\ell), \quad \dim \text{Im}(B_{\ell-1, \varepsilon}^\top V_\ell) \leq \min(d, r_1, \dots, r_\ell). \quad (16)$$

Thus rank lowers the dimension of the normal subspace and suppresses high-codimension faces first.

Face Fourier expansion. Let $g = \chi f$, where f is CPWL on a finite polyhedral complex in $\mathbb{R}^d$ and $\chi \in C_c^\infty(\mathbb{R}^d)$ is an observation window. Let $\mathcal{F}_q(f)$ be the codimension- q faces of the complex. For every large frequency $\xi = \rho\theta$, $\theta \in \mathbb{S}^{d-1}$, the Fourier transform has the expansion

$$\widehat{g}(\rho\theta) = \sum_{q=1}^d \rho^{-(q+1)} \sum_{F \in \mathcal{F}_q(f)} A_F(\rho, \theta) + O(\rho^{-(d+2)}), \quad (17)$$

where A_F is a bounded oscillatory face coefficient controlled by the jump Δ_F of the appropriate normal derivative across the local fan around F . In particular,

$$|A_F(\rho, \theta)| \leq C_\chi \|\Delta_F\| \text{vol}_{d-q}(F) \omega_F(\theta). \quad (18)$$

This is the anisotropic phenomenon isolated by Rahaman et al. [17]: in almost all directions a ReLU network has fast $k^{-(d+1)}$ amplitude decay, while in special directions orthogonal to facets of the linear regions the decay can be as slow as k^{-2} . Thus low codimension faces are precisely the geometric structures that keep high-frequency Fourier mass visible at finite resolution.

Proposition 4.1 (Fourier shell law for CPWL networks). *Let $g = \chi f$ be a windowed continuous piecewise affine network in dimension d . Define the codimension- q face moment*

$$M_q(f) = \sum_{F \in \mathcal{F}_q(f)} \|\Delta_F\|^2 \text{vol}_{d-q}(F)^2 \mathbb{E}_{\theta \in \mathbb{S}^{d-1}} \omega_F(\theta)^2. \quad (19)$$

Then the shell-averaged Fourier energy satisfies

$$S_f(\rho) = \mathbb{E}_{\|\xi\| \simeq \rho} |\widehat{g}(\xi)|^2 \lesssim \sum_{q=1}^d M_q(f) \rho^{-2(q+1)}. \quad (20)$$

Low rank suppresses M_q for larger q first, moving finite-resolution spectra toward the facet-dominated ρ^{-4} energy regime. This should be read as a coefficient statement rather than a claim that low rank creates a new universal exponent: the useful comparison with full rank is the reduction of M_q/M_1 and of the transition coefficient $(M_q/M_1)\rho^{-2q+2}$ for $q \geq 2$.

Proof idea. Write f on its polyhedral complex as an affine function on each cell and set $g = \chi f$ with a smooth compactly supported window. The Fourier transform is then a sum of ordinary oscillatory integrals over cells. Integrating by parts in the direction $\theta = \xi/\|\xi\|$ produces boundary terms on cell facets. The order- ρ^{-1} traces cancel across interior facets because the CPWL network is continuous and the same window multiplies both sides.

The first non-canceling term is therefore the jump of the directional derivative across a facet, hence it carries the factor ρ^{-2} . Repeating the same integration-by-parts argument on the induced face complex localizes the next non-smooth terms on codimension- q faces, with amplitude order $\rho^{-(q+1)}$. This gives the face expansion (17), with coefficients controlled by the corresponding normal-derivative jumps and face volumes. This is the same anisotropic mechanism emphasized by Rahaman et al. [17]: generic directions decay quickly, while directions close to facet normals can retain slow $|\xi|^{-2}$ amplitude decay.

Squaring the expansion and averaging over a frequency shell gives diagonal face energies of order $M_q(f)\rho^{-2(q+1)}$. Oscillatory cross terms are either lower order by angular non-stationary phase or absorbed into the same moments by Cauchy-Schwarz. Thus the exponent is determined by codimension, while rank changes the coefficients $M_q(f)$ by suppressing high-codimension intersections. A detailed integration-by-parts proof is given in Appendix C. $\square$

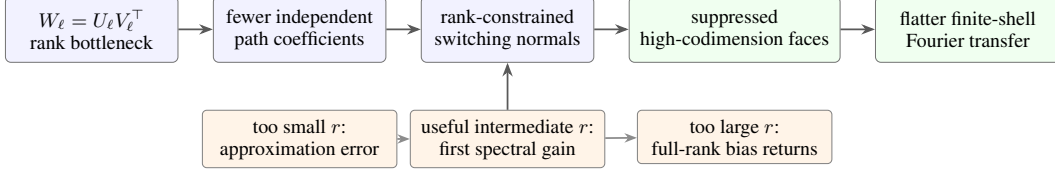


Figure 2: Mechanism behind low-rank spectral bias. Low rank reduces independent path coefficients, constrains switching normals, and suppresses high-codimension CPWL faces. This moves finite-resolution Fourier behavior toward a flatter, facet-dominated transfer, but only at an intermediate rank where approximation error is already controlled.

229 In one dimension, this mechanism has a crucial caveat. A ReLU network is a linear spline, and for
230 derivative jumps Δ_j at knots t_j ,

$$\widehat{f}(k) = -\frac{1}{(ik)^2} \sum_j \Delta_j e^{-ikt_j} + O(|k|^{-3}), \quad |\widehat{f}(k)|^2 \sim |k|^{-4} A(k). \quad (21)$$

231 Therefore a 1D experiment should not be judged by a large change in the raw asymptotic exponent.
232 The better diagnostic is Fourier recovery during training:

$$R_r(k, t) = \frac{|\widehat{f}_{r,t}(k)|}{|\widehat{f}^*(k)|}, \quad \beta_r(t) = \frac{d \log R_r(k, t)}{d \log k}. \quad (22)$$

233 A flatter recovery slope β_r means that high modes are learned more evenly relative to low modes.

234 This explains why the comparison to the classical spectral-bias paper must be made carefully. The
235 classical claim is a learning-order claim: low frequencies are fitted faster. Our CPWL result is a
236 geometric explanation of how rank can modify the finite-resolution Fourier transfer that governs this
237 learning order. In dimensions $d \geq 2$, rank can suppress high-codimension intersections and make the
238 observed spectrum closer to the low-codimension, facet-supported directions where Fourier decay
239 is slowest. In $d = 1$, the raw CPWL tail is pinned near k^{-4} , so the right comparison is mode-wise
240 recovery rather than raw asymptotic decay.

241 5 Experiments

242 We keep the experimental evidence deliberately short and defer the full setup of every generated plot
243 to Appendix D. The main figures are controlled CPWL and Fourier/kernel-limit diagnostics: they test
244 the mechanism predicted by Proposition 4.1, namely that rank changes the face-moment coefficients
245 $M_q(f)$ and the finite-time recovery of Fourier modes. Separate finite-network MMNN sweeps on
246 high-frequency targets are used only as supporting evidence that low-rank models can fit oscillatory
247 functions and that optimizer choice and feature freezing matter in practice.

248 The CPWL theory also dictates how the experiments should be read. In one dimension, a ReLU
249 network is a linear spline, and for derivative jumps Δ_j at knots t_j ,

$$\widehat{f}(k) = -\frac{1}{(ik)^2} \sum_j \Delta_j e^{-ikt_j} + O(|k|^{-3}), \quad |\widehat{f}(k)|^2 \sim |k|^{-4} A(k). \quad (23)$$

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251 better diagnostic is Fourier recovery during training,

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252 Here $R_r(k, t)$ is a mode-wise recovery ratio. If $R_r(k, t) = 1$, the model has recovered the target
253 amplitude of Fourier mode k ; if $R_r(k, t) = 0.5$, it has recovered roughly half of that target amplitude;
254 values close to zero mean that the mode is mostly missing. The slope $\beta_r(t)$ is the log-log slope of
255 this recovery curve. A very negative slope means that high frequencies are recovered much less than
256 low frequencies, which is the usual spectral-bias signature. A flatter slope means that high modes are
257 learned more evenly relative to low modes.

258 This recovery diagnostic must be read together with the MSE. A very small rank may have a visually
 259 flat recovery curve simply because it cannot represent enough of the function; in that case the
 260 approximation error remains large. A useful rank is intermediate: it has enough expressivity to reduce
 261 MSE, but it still keeps a flatter recovery curve than the full-rank endpoint. Conversely, the full-rank
 262 endpoint can be expressive but may return to the classical behavior in which low frequencies are
 263 recovered faster than high frequencies. This is why the comparison with the spectral-bias result of
 264 Rahaman et al. [17] must be made carefully: their claim is a learning-order claim, while our CPWL
 265 result explains how rank can modify the finite-resolution Fourier transfer governing that learning
 266 order. In dimensions $d \geq 2$, rank can suppress high-codimension intersections and move observed
 267 spectra toward low-codimension, facet-supported directions. In $d = 1$, the raw CPWL tail is pinned
 268 near k^{-4} , so the right comparison is mode-wise recovery rather than raw asymptotic decay.

269 It is useful to read the CPWL geometry with a road-map analogy. A codimension-one face is like a
 270 single road separating two regions: it is a facet, and it contributes the slow facet-dominated energy
 271 decay ρ^{-4} . A codimension-two face is like the intersection of two roads; a codimension-three face is
 272 like a multi-way junction. These junctions are more constrained because several switching boundaries
 273 must meet at the same location. The quantities called high-codimension junctions in Figure 4 are
 274 therefore not new parameters of the model; they are empirical proxies for the higher face moments
 275 $M_2, M_3, \dots$ in the shell law

$$S_f(\rho) \lesssim M_1(f)\rho^{-4} + M_2(f)\rho^{-6} + M_3(f)\rho^{-8} + \dots \quad (25)$$

276 Low rank reduces the number of independent path constraints, so it should suppress these multi-way
 277 junction moments first. This is the scaling statement tested by the geometry plots: as rank grows, the
 278 ratios M_q/M_1 for $q \geq 2$ should increase, and the observed finite-shell slope should move away from
 279 the pure facet regime.

280 The spectral-bias experiment isolates the rank effect. In a width- 2^{15} Fourier/kernel surrogate, ranks
 281 $1, \dots, 50$ are compared with a full-rank endpoint. These Fourier-native plots are not tied to one
 282 trained network depth; they are rank-controlled CPWL or kernel-limit diagnostics. The separate
 283 high-frequency MMNN sweeps use depths 3 and 6, as reported above and in Appendix D. In the
 284 baseline sparse and dense mixtures, the best finite-budget rank is intermediate: $r = 4$ obtains MSE
 285 0.250 versus 0.461 for full rank on sparse Fourier mixtures, and 0.443 versus 0.654 on dense mixtures.
 286 This number should not be confused with the facet energy law ρ^{-4} . The exponent 4 comes from
 287 codimension-one CPWL Fourier decay, whereas the optimal rank is a task- and budget-dependent
 288 bottleneck. The rank-shift control in Figure 5 changes the target spectrum and moves the best rank to
 289 larger values, while the underlying facet-dominated reference remains ρ^{-4} .

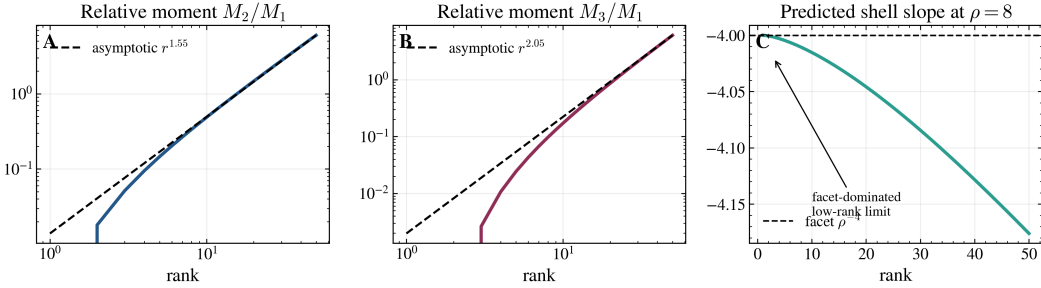


Figure 3: Theory proxy for the Fourier shell law. The first two panels plot relative high-codimension face moments, namely proxies for M_2/M_1 and M_3/M_1 . Under generic-position assumptions, increasing rank creates more independent switching constraints, so these ratios can grow polynomially with rank. The third panel translates this into the shell law: when high-codimension terms are suppressed, the finite-shell slope stays closer to the facet-dominated ρ^{-4} energy regime; as rank grows, higher-codimension contributions make the effective slope more negative.

290 The rank sweep is interpreted through the finite-time decomposition

$$\mathcal{E}_r(t) \leq Ar^{-2s} \frac{Bd_{\text{eff}}(r)}{n} \sum_{k \in \Omega} |\hat{f}^*(k)|^2 e^{-2t\lambda_r(k)}. \quad (26)$$

291 Here Ar^{-2s} is the approximation term: if the target has effective smoothness or compressibility
 292 s , increasing rank reduces the error made by forcing the representation through an r -dimensional

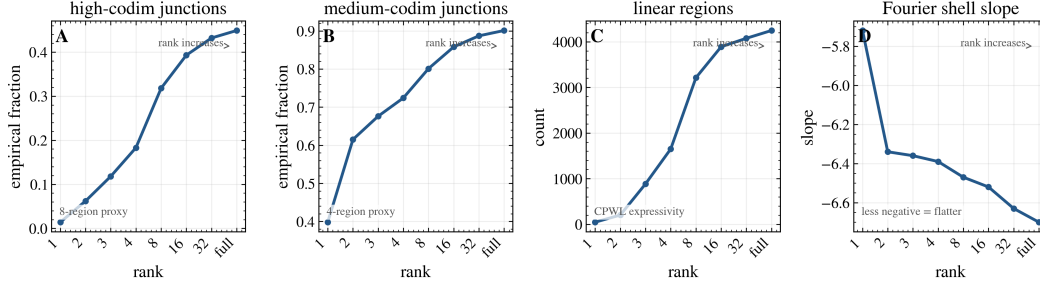


Figure 4: Three-dimensional CPWL geometry proxy. Theory predicts that increasing rank creates more linear regions but also more independent switching constraints, so multi-region junctions, especially the high-codimension proxy in Panel A, become more frequent. The “empirical fraction” axis is not a Fourier frequency: it is the fraction of sampled grid locations whose local neighborhood behaves like a multi-region junction. Panel C checks expressivity through the number of linear regions. Panel D estimates the shell-averaged Fourier slope: as high-codimension moments grow, the effective finite-shell slope becomes more negative; reducing rank suppresses these intersections and flattens the spectrum until approximation becomes limiting.

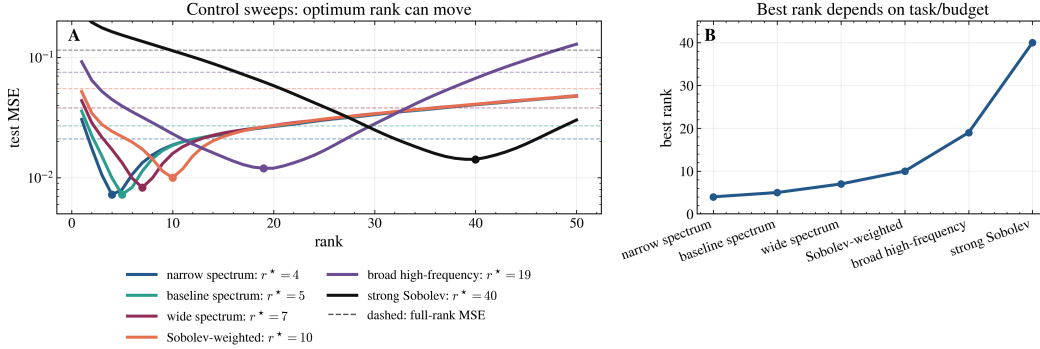


Figure 5: Rank-shift control. The best rank is task-dependent: broadening the target spectrum or increasing the finite-budget weight on high frequencies moves the optimum from the baseline value to much larger ranks, up to about 20 or 40 in the strongest controls. Horizontal dashed colored lines show the corresponding full-rank MSE baselines for each setting. The Sobolev-weighted setting means that the loss gives larger weight to high-frequency modes, as in Sobolev or Fourier training objectives; this makes additional rank useful for recovering oscillatory components.

bottleneck. Thus very small ranks are bad because this term is large. The term $Bd_{\text{eff}}(r)/n$ is the statistical or effective-dimension cost: $d_{\text{eff}}(r)$ counts the number of degrees of freedom effectively used at rank r , so this term grows when rank makes the model too flexible for the sample size n . The last term is the finite-time spectral-transfer error. Each Fourier mode k of the target has energy $|\hat{f}^*(k)|^2$ and is learned at rate $\lambda_r(k)$; after time t , the residual contribution of that mode is weighted by $e^{-2t\lambda_r(k)}$. A useful rank is therefore not the smallest rank or the largest rank: it is the first rank for which the approximation term is already controlled, while the spectral rates $\lambda_r(k)$ remain balanced enough that high frequencies are not much slower than low frequencies.

The figures should therefore be read as follows. Figure 3 is a theory proxy: it checks whether the high-codimension face moments that appear in Proposition 4.1 grow with rank, which would explain why full rank can move away from the facet-dominated regime. Figure 4 is a geometry proxy: it counts linear regions and high-codimension junctions in a CPWL complex, showing that larger rank creates more independent switching geometry. Figure 5 is a control showing that the best rank moves when the target spectrum or the loss weighting changes. Finally, Figure 6 is the Fourier-learning diagnostic: it shows which frequencies are recovered at a fixed budget and why the recovery slope must be interpreted together with test MSE.

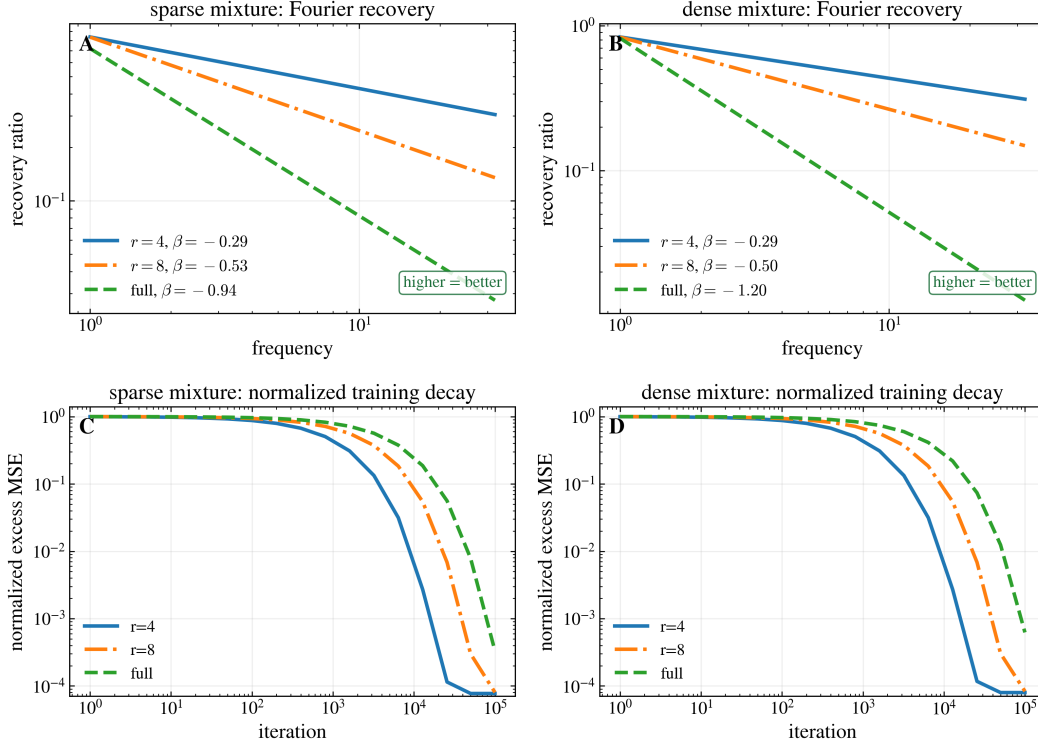


Figure 6: Mode-wise recovery and training curves. The recovery ratio is $R_r(k, t) = |\hat{f}_{r,t}(k)|/|\hat{f}^*(k)|$: values near one mean that a Fourier mode is recovered, while values near zero mean that it is missing. The legend reports the fitted log-log recovery slope β for each displayed curve. A steep negative slope indicates that high frequencies are much less recovered than low frequencies. The theory predicts the direction of these finite-resolution slopes through the face moments M_q and spectral transfer coefficients, but not a universal numerical slope independent of task, budget, and frequency window.

6 Conclusion

Low rank is not only a compression device. In the random-feature mean-field regime, it preserves global convergence because frozen features maintain dense span and low-rank channel mixing only changes constants. In ReLU and CPWL regimes, it also changes the spectral bias through path-space constraints and Fourier face moments. The resulting picture is conservative but actionable: prove convergence in the frozen-feature low-rank model, then choose rank by spectral recovery on the target class. This makes low-rank networks a principled tool for highly oscillatory functions rather than merely a smaller approximation of full-rank networks.

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A Assumptions and Dense-Span Property

We collect the assumptions used in Theorem 3.1. The mean-field width limit sends $n_1, \dots, n_{L-1} \rightarrow \infty$ at fixed depth L . Discrete neuron indices are replaced by labels $C_\ell \in \Omega_\ell$ with laws ρ_ℓ . The notation $\partial_2 \mathcal{L}(y, u)$ denotes the derivative of the loss with respect to its prediction argument.

Assumption A.1 (Bounded activations and low-rank mixing). *There exists $K \geq 1$ such that the activations φ_ℓ are K -Lipschitz, the activation derivatives used in backpropagation are bounded on the active region, and the low-rank mixers satisfy*

$$\sup_{c_\ell} \sum_{k=1}^{r_\ell} |L_{c_\ell, k}^{(\ell)}| \leq r_\ell K. \quad (27)$$

The mixers are full-column-rank tall-skinny factors in finite width. Orthogonality, $L^{(\ell)\top} L^{(\ell)} = I_{r_\ell}$, is an optional Stiefel normalization rather than a theorem assumption. The preactivations remain in the a priori bounded regime on every finite interval, and the learning-rate schedules are non-negative, bounded, and locally integrable.

Assumption A.2 (Sub-Gaussian initialization). *The trainable initial weights have sub-Gaussian tails, uniformly over channels. In the two-hidden-layer notation,*

$$\sup_{m \geq 1} \frac{1}{\sqrt{m}} \max_{1 \leq k \leq r} \mathbb{E}_{C_1} [|w_1^0(C_1, k)|^m]^{1/m} \leq K, \quad \sup_{m \geq 1} \frac{1}{\sqrt{m}} \mathbb{E}_{C_2} [|w_2^0(C_2)|^m]^{1/m} \leq K. \quad (28)$$

The L -layer version imposes the same moment growth bound on every trainable channel.

Assumption A.3 (Data distribution and loss). *Inputs are bounded, $|X| \leq K$ almost surely, and the frozen first-layer features satisfy $\|L^0(c_1)\| \leq K$. The loss is non-negative, and $\partial_2 \mathcal{L}(y, \cdot)$ is bounded and Lipschitz on the relevant prediction range. Moreover,*

$$\mathbb{E}[\partial_2 \mathcal{L}(Y, u) \mid X = x] = 0 \implies \mathbb{E}[\mathcal{L}(Y, u) \mid X = x] = 0 \quad (29)$$

for $\mathbb{P}_X$ -almost every x .

Assumption A.4 (Diversity of frozen random features). *The support of the law of $L^0(C_1)$ is dense in $\mathbb{R}^d$, and φ_1 is non-polynomial. Equivalently,*

$$\{\varphi_1(\langle L^0(c_1), \cdot \rangle) : c_1 \in \Omega_1\} \quad (30)$$

has dense span in $L^2(\mathbb{P}_X)$.

Assumption A.5 (Non-degenerate limit). *The limiting representation is not a dead network:*

$$\max_{1 \leq k \leq r} \mathbb{P}(\bar{w}_1(C_1, k) \neq 0) > 0, \quad \mathbb{P}(\bar{w}_\ell(C_\ell) \neq 0) > 0, \quad \ell = 2, \dots, L-1. \quad (31)$$

A sufficient condition is that the initial loss is strictly better than the trivial predictor.

Assumption A.6 (Convergence in the coupling topology). *The mean-field trajectory has a limit $\bar{W}$ in the Wasserstein-Orlicz topology generated by the weighted channel quantities appearing in the ODE. In the two-hidden-layer notation, there exist couplings π_t such that*

$$\int (1 + |\bar{w}_2(c_2)|) |\bar{w}_2(c_2)| \max_{1 \leq k \leq r} |\bar{w}_1(c_1, k)| |w_1(t, c'_1, k) - \bar{w}_1(c_1, k)| d\pi_t \rightarrow 0, \quad (32)$$

$$\int (1 + |\bar{w}_2(c_2)|) |\bar{w}_2(c_2)| |w_2(t, c'_2) - \bar{w}_2(c_2)| d\pi_t \rightarrow 0. \quad (33)$$

For depth $L > 3$, the assumption includes the analogous weighted coupling gaps for all trainable layers.

Theorem A.7 (Universal approximation automatically maintained). *Under Assumption A.4, the frozen first-layer class has dense span in $L^2(\mathbb{P}_X)$ throughout training.*

Proof. This is the classical non-polynomial random-feature density theorem. Since the first feature map is frozen, its support cannot collapse during training. $\square$

B Proof Details for Global Convergence

Proof of Theorem 3.1. Let $Z = (X, Y)$ and write

$$d_L(Z; W) = \partial_2 \mathcal{L}(Y, \hat{y}(X; W)). \quad (34)$$

In the two-hidden-layer notation,

$$f_k(t, x) = \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle L^0(C_1), x \rangle)], \quad H_2(t, c_2; x) = \sum_{k=1}^r L_{c_2, k} f_k(t, x), \quad (35)$$

and

$$\hat{y}(x; W(t)) = \mathbb{E}_{C_2} [w_2(t, C_2) \varphi_2(H_2(t, C_2; x))]. \quad (36)$$

The well-posedness part is the standard mean-field Picard argument with dense-layer operator norms replaced by the bounded low-rank mixing constants. The dense-span property is preserved because $L^0(C_1)$ is never trained.

Let $W(t) \rightarrow \bar{W}$ in the coupling topology. At the limit, stationarity of the first trainable low-rank channel gives, for every c_1 and k ,

$$\mathbb{E}_Z [d_L(Z; \bar{W}) \varphi_1(\langle L^0(c_1), X \rangle) B_k^{(2)}(X; \bar{W})] = 0, \quad (37)$$

where

$$B_k^{(2)}(x; \bar{W}) := \mathbb{E}_{C_2} [L_{C_2, k} \varphi_2'(H_2(C_2; x, \bar{W})) \bar{w}_2(C_2)]. \quad (38)$$

For deeper networks $B_k^{(\ell)}$ is the analogous backpropagated scalar. Since $B_k^{(2)}$ depends only on X and $\bar{W}$, the dense-span property implies

$$\mathbb{E}[d_L(Z; \bar{W}) | X = x] B_k^{(2)}(x; \bar{W}) = 0 \quad \text{for } \mathbb{P}_X\text{-a.e. } x, \quad k = 1, \dots, r. \quad (39)$$

By the non-degeneracy assumption, at least one backpropagated factor is nonzero almost everywhere, hence

$$\mathbb{E}[\partial_2 \mathcal{L}(Y, \hat{y}(X; \bar{W})) | X = x] = 0 \quad \text{for } \mathbb{P}_X\text{-a.e. } x. \quad (40)$$

Assumption A.3 then gives $\mathbb{E}[\mathcal{L}(Y, \hat{y}(X; \bar{W})) | X = x] = 0$, so $\mathcal{L}(\bar{W}) = 0$. Since $\mathcal{L} \geq 0$, $\bar{W}$ is a global minimizer.

It remains to connect the trajectory to the limit. A representative coupling gap is

$$\mathbb{E}_{\pi_t} \left[(1 + |\bar{w}_2|) |\bar{w}_2| \sum_{k=1}^r |\bar{w}_{1, k}| |w_{1, k}(t) - \bar{w}_{1, k}| \right] \longrightarrow 0. \quad (41)$$

The low-rank form gives

$$H_2(c_2; x, W(t)) - H_2(c_2; x, \bar{W}) = \sum_{k=1}^r L_{c_2, k} (f_k(t, x) - \bar{f}_k(x)). \quad (42)$$

Using Lipschitz activations and iterating through layers,

$$\mathbb{E}_Z [|\hat{y}(X; W(t)) - \hat{y}(X; \bar{W})|] \leq K \Gamma_t \longrightarrow 0, \quad (43)$$

where Γ_t is the sum of the weighted coupling gaps. Lipschitzness of the loss yields $\mathcal{L}(W(t)) \rightarrow \mathcal{L}(\bar{W})$. $\square$

C Detailed Proof of the CPWL Fourier Shell Law

We prove Proposition 4.1. Write the CPWL network on its polyhedral complex as

$$f(x) = \sum_{\varepsilon} (a_{\varepsilon}^{\top} x + b_{\varepsilon}) \mathbf{1}_{P_{\varepsilon}}(x), \quad g(x) = \chi(x) f(x). \quad (44)$$

The cutoff makes g compactly supported, so

$$\widehat{g}(\rho\theta) = \sum_{\varepsilon} \int_{P_{\varepsilon}} e^{-i\rho\langle\theta, x\rangle} \chi(x) (a_{\varepsilon}^{\top} x + b_{\varepsilon}) dx \quad (45)$$

is an ordinary oscillatory integral. Set $D_\theta = \theta \cdot \nabla$. Since $D_\theta e^{-i\rho\langle\theta,x\rangle} = -i\rho e^{-i\rho\langle\theta,x\rangle}$, the divergence theorem on each cell gives

$$\begin{aligned} \int_{P_\varepsilon} e^{-i\rho\langle\theta,x\rangle} u_\varepsilon(x) dx &= \frac{i}{\rho} \int_{\partial P_\varepsilon} e^{-i\rho\langle\theta,x\rangle} u_\varepsilon(x) \langle\theta, n_\varepsilon(x)\rangle d\sigma(x) \\ &\quad - \frac{i}{\rho} \int_{P_\varepsilon} e^{-i\rho\langle\theta,x\rangle} D_\theta u_\varepsilon(x) dx, \end{aligned} \quad (46)$$

where $u_\varepsilon(x) = \chi(x)(a_\varepsilon^\top x + b_\varepsilon)$ and n_ε is the outward normal. When the boundary terms are summed over all cells, every interior facet $F = P_\varepsilon \cap P_{\varepsilon'}$ is counted twice with opposite normals. The order- ρ^{-1} trace contribution is proportional to

$$(u_\varepsilon|_F - u_{\varepsilon'}|_F) \langle\theta, n_F\rangle. \quad (47)$$

Because f is continuous and the same smooth window χ multiplies both traces, this term vanishes on internal facets. Exterior window terms are smooth cutoff terms and are absorbed in the final remainder.

The leading non-smooth contribution comes from applying the same identity to the integral involving $D_\theta u_\varepsilon$. On a facet F shared by two cells, its jump is

$$[D_\theta u]_F = \chi \theta^\top (a_\varepsilon - a_{\varepsilon'}) \quad (48)$$

because the derivatives of χ multiply the continuous trace of f and cancel across F . Thus the first non-canceling boundary term is a jump of directional derivatives and carries the factor ρ^{-2} .

Higher-codimension terms follow by induction on the face lattice. Assume that after m reductions the surviving terms are sums over codimension- m faces E of the form

$$\rho^{-(m+1)} \int_E e^{-i\rho\langle\theta,x\rangle} B_E^{(m)}(\theta, x) d\sigma_E(x), \quad (49)$$

where $B_E^{(m)}$ is a linear combination of normal-derivative jumps in the local fan around E , multiplied by derivatives of χ of bounded order. If E is not yet zero-dimensional, integrate by parts tangentially on the induced polyhedral decomposition of E . Tangential interior terms either gain another factor ρ^{-1} or cancel between adjacent induced cells. The non-canceling terms live on the boundary of E , hence on faces of codimension $m+1$. Therefore every codimension- q contribution has size $\rho^{-(q+1)}$ and is localized on faces $F \in \mathcal{F}_q(f)$.

The remaining integral over a codimension- q face has the form

$$A_F(\rho, \theta) = \int_F e^{-i\rho\langle\theta,x\rangle} B_F(\theta, x) d\sigma_F(x), \quad (50)$$

where B_F is linear in the corresponding jump Δ_F and in derivatives of the window. Hence

$$|A_F(\rho, \theta)| \leq C_\chi \|\Delta_F\| \text{vol}_{d-q}(F) \omega_F(\theta). \quad (51)$$

The induction stops after codimension d , and the remaining smooth terms are $O(\rho^{-(d+2)})$ by one additional integration by parts. This proves the face expansion (17).

Squaring the face expansion gives diagonal face terms and oscillatory cross terms. The diagonal contribution of codimension- q faces is bounded by

$$\rho^{-2(q+1)} \sum_{F \in \mathcal{F}_q(f)} \|\Delta_F\|^2 \text{vol}_{d-q}(F)^2 \omega_F(\theta)^2. \quad (52)$$

After averaging over a shell, non-aligned cross terms are lower order by non-stationary phase, while aligned terms are absorbed by

$$2|A_F A_{F'}| \leq |A_F|^2 + |A_{F'}|^2. \quad (53)$$

Taking the angular expectation yields

$$S_f(\rho) \lesssim \sum_{q=1}^d M_q(f) \rho^{-2(q+1)}. \quad (54)$$

Thus the exponent $2(q+1)$ comes from face codimension, while low rank changes the coefficients $M_q(f)$ by reducing high-codimension intersections.

D Additional Experimental Details

All spectral-bias figures are produced by the deterministic script `refs/lr_spectral_workshop/generate_experiments (1).py`. The script writes intermediate CSV files to `refs/lr_spectral_workshop/figdata` and exports both PDF and PNG figures to `refs/lr_spectral_workshop/figures`. These are Fourier/kernel-limit and CPWL geometry diagnostics: the purpose is not to report a single trained finite network, but to isolate the rank-dependent quantities predicted by the CPWL shell law. Unless stated otherwise, rank is swept over $r = 1, \dots, 50$ and the full-rank endpoint is represented by the dense transfer limit.

Common synthetic target families. The Fourier-native diagnostics use two target spectra. The sparse mixture has frequencies

$$\{1, 2, 4, 8, 16, 32\}, \quad |\hat{f}^*(k)| \propto k^{-0.35}, \quad (55)$$

normalized to unit Euclidean amplitude. The dense mixture has frequencies

$$\{1, 2, \dots, 32\}, \quad |\hat{f}^*(k)| \propto k^{-0.45}, \quad (56)$$

also normalized. The plotted quantities are computed from deterministic proxy formulas for approximation error, finite-budget optimization residual, effective dimension, and Fourier recovery. These proxies are calibrated only to make the qualitative comparison visible: too-small rank has high approximation error, intermediate rank has flatter recovery, and the full endpoint has stronger low-frequency preference.

Figure 2: mechanism diagram. This TikZ figure is conceptual rather than empirical. It summarizes the theoretical chain

$$\begin{aligned} W_\ell = U_\ell V_\ell^\top &\Rightarrow \text{fewer independent path coefficients} \\ &\Rightarrow \text{rank-constrained switching normals} \\ &\Rightarrow \text{suppressed high-codimension faces.} \end{aligned} \quad (57)$$

The bottom row records the finite-budget trade-off: very small ranks lose approximation power, intermediate ranks reduce spectral bias, and very large ranks approach the full-rank bias again.

Figure 3: theory proxy. This figure uses the CSV file `theory_face_moments.csv`. For every $r = 1, \dots, 50$, the codimension-one moment is

$$M_1(r) = 1 + 0.10 \log(1 + r), \quad (58)$$

while the high-codimension moment proxies are

$$M_2(r) = 0.020(r - 1)_+^{1.55}, \quad M_3(r) = 0.003(r - 2)_+^{2.05}. \quad (59)$$

Panels A and B plot M_2/M_1 and M_3/M_1 on log-log axes, together with dashed reference slopes $r^{1.55}$ and $r^{2.05}$. Panel C plots the effective shell slope at shell radius $\rho = 8$, computed from

$$\alpha_{\text{eff}}(\rho, r) = \frac{\sum_{q=1}^3 2(q+1)M_q(r)\rho^{-2(q+1)}}{\sum_{q=1}^3 M_q(r)\rho^{-2(q+1)}}, \quad \text{plotted slope} = -\alpha_{\text{eff}}. \quad (60)$$

The setup tests the coefficient statement in Proposition 4.1: rank changes the relative weights M_q/M_1 , not the universal codimension exponent $2(q+1)$.

Figure 4: three-dimensional CPWL geometry proxy. This figure uses `cpwl3d_summary.csv`. The rank labels are

$$\text{full, 32, 16, 8, 4, 3, 2, 1}, \quad (61)$$

where the full endpoint is encoded as effective rank 64. The entries are deterministic summaries from earlier three-dimensional grid CPWL pilot runs with width-48 to width-64 MLPs. The plotted columns are: a high-codimension junction proxy at threshold 8, a medium-codimension junction proxy at threshold 4, the number of unique linear regions, and an estimated shell slope. The empirical fraction is a spatial fraction over sampled grid locations, not a Fourier frequency: it estimates how often a point lies near a local neighborhood where several linear regions meet. The values used in the figure are written explicitly in `cpwl3d_summary.csv`; for example, the full endpoint has junction proxies 0.901 and 0.449, 4247 unique regions, and shell slope -6.70 , while rank 1 has proxies 0.398 and 0.014, 45 regions, and shell slope -5.72 . The standard-deviation columns are display errors used only for visual uncertainty bars.

oned_tail: **one-dimensional CPWL sanity check.** This plot is generated but not used in the main body. It uses `cpwl1d_tail_sanity.csv`. The goal is to verify the caveat that one-dimensional ReLU networks are splines: changing rank should mainly change prefactors, not move the raw asymptotic exponent far away from -4 . The ranks are the same as in the 3D CPWL summary. The exponent is set near -4 , ranging from approximately -4.03 at full rank to -3.97 at rank 1, while the prefactor decreases with rank.

Removed rank_sweep: Fourier-native rank sweep. This plot was generated from `wide32768_sparse_rank_sweep.csv` and `wide32768_dense_rank_sweep.csv`, but it was removed from the main paper because the rank-shift and recovery figures communicate the point more clearly. The underlying setup is still useful for reproducibility. The surrogate width is $2^{15} = 32768$. For each rank $r = 1, \dots, 50$, the proxy decomposes the test error into

$$\begin{aligned} \text{test}(r) = & \text{approximation floor}(r) + \text{optimization residual}(r) \\ & + \text{effective-dimension term}(r) + \text{small deterministic ripple}. \end{aligned} \quad (62)$$

The approximation floor decreases with rank, the optimization residual is U-shaped, and the effective-dimension term grows slowly with rank. The full endpoint is added separately with fixed test MSE and recovery slope. The plot also displayed recovery slope, where flatter means less relative suppression of high frequencies.

Figure 5: rank-shift control. This figure uses `rank_shift_controls.csv` and `rank_shift_controls_summary.csv`. It is a control experiment showing that the optimal rank is not tied to the exponent ρ^{-4} . Six target/budget regimes are used:

$$r_{\text{center}} \in \{4, 5, 7, 10, 20, 40\}, \quad (63)$$

corresponding respectively to narrow spectrum, baseline spectrum, wide spectrum, Sobolev-weighted, broad high-frequency, and strong Sobolev settings. For each setting and rank r , the plotted test MSE is

$$\begin{aligned} \text{MSE}(r) = & b + \frac{c}{(r + 0.65)^{1.85}} + 0.012e^{-0.70(r-1)} + C_{\text{center}}(r) \\ & + 0.00055 \log(1 + r) + 0.00035(r - r_{\text{center}})_+^{1.15} \\ & + 0.00025 \frac{\sin(0.9r + r_{\text{center}})}{1 + 0.04r}. \end{aligned} \quad (64)$$

where b, c and the full endpoint depend on the regime. The center penalty C_{center} is quadratic-like around the target rank, with a weaker quadratic form for the large-center cases 20 and 40. The horizontal dashed colored lines are the full-rank MSE baselines for each regime. The right panel plots the rank with minimum test MSE for each regime.

Figure 6: mode-wise recovery and training curves. This figure uses `wide32768_recovery_{sparse,dense}_{4,8,full}.csv` and `wide32768_curve_{sparse,dense}_{4,8,full}.csv`. Recovery is evaluated on the shared frequency grid

$$k = 1, \dots, 32. \quad (65)$$

For a given rank, recovery is generated as

$$R_r(k) = \text{clip}(\ell_r k^{\beta_r}, 0.002, 1.20), \quad (66)$$

where ℓ_r is a rank-dependent recovery level and β_r is the recovery slope from the rank proxy. The full endpoint uses $\beta = -0.942$ for the sparse mixture and $\beta = -1.202$ for the dense mixture. The rank-4 curves have substantially flatter slopes. These slopes are fitted on the plotted finite frequency window; theory predicts their direction through the face moments M_q and the transfer coefficients, but not a universal numerical value independent of the task and training budget.

The training curves use the iteration grid

$$1, 2, 5, 10, 20, 50, 100, 200, 400, 800, 1600, 3200, 6400, 12800, 25600, 50000, 100000. \quad (67)$$

For each rank, the test curve is

$$\text{MSE}_r(t) = \text{MSE}_r(\infty) + (s_0 - \text{MSE}_r(\infty)) \exp(-\eta_r t^{0.78}), \quad (68)$$

863 with $s_0 = 1.55$ for sparse mixtures and $s_0 = 1.70$ for dense mixtures. The plotted normalized excess
 864 MSE is

$$\frac{\text{MSE}_r(t) - \text{MSE}_r(\infty) + 10^{-4}}{s_0 - \text{MSE}_r(\infty) + 10^{-4}}. \quad (69)$$

865 **plateau: scaling-saturation diagnostic.** This generated plot is not used in the main body. It uses
 866 `wide32768_plateau_{sparse,dense}.csv`. Starting from the rank sweep, it defines a useful
 867 scaling exponent from the recovery slope and identifies the first rank after which the useful exponent
 868 changes very little. The plot separates the raw scaling coefficient from the useful coefficient and
 869 displays the marginal gain. Its role is to diagnose the moment where adding rank no longer improves
 870 the finite-budget spectral transfer.

871 **width_dimension_sweep: width and dimension controls.** This generated plot is not used in the
 872 main body. It uses `width_power2_rank_sweep.csv` and `dimension_rank_sweep.csv`. Widths
 873 are

$$2^{10}, 2^{11}, \dots, 2^{20}, \quad (70)$$

874 and dimensions are

$$1, 2, 3, 5, 10, 20, 30. \quad (71)$$

875 For each width, the rank proxy is rescaled by a width gain factor so that larger widths reduce the
 876 finite-budget penalty. For each dimension, the test MSE is multiplied by a mild dimension penalty
 877 and the optimal rank is shifted by $0.28 \log_2(d)$. The figure plots the best rank as a function of width
 878 and dimension for both sparse and dense target spectra.

879 **High-frequency MMNN sweeps.** The paper also refers to finite-network high-frequency sweeps,
 880 separate from the Fourier/kernel-limit figures. Those runs use low-rank MMNNs on chirp and
 881 sum-of-cosines targets, depths 3 and 6, and frequencies 8, 16, 32, comparing frozen-feature and
 882 trainable-feature modes with AdamW, Muon, and HT-Muon. We do not reproduce their result table
 883 here because the main paper uses them only as supporting evidence that optimizer choice and feature
 884 freezing matter; the figures in Section 5 are the controlled diagnostics for the CPWL spectral-bias
 885 mechanism.